



Grade K Science

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · KINDERGARTEN · FULL YEAR

160 min per week · 2 lessons weekly

About this overview. This document is an example of how the course might be taught. This overview is drawn from the Florida course description and Optima's 2026–27 course plan rather than a fully built course, so scholars may see some variation in investigations, examples, and projects from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

In Kindergarten Science, scholars explore the natural world through observation, investigation, and inquiry, discovering patterns in Earth, life, and physical science such as day and night cycles, the five senses, motion, and the characteristics of living things. The course cultivates wonder, collaboration, and sensory-based exploration as scholars pose questions, make observations with their senses, and record what they learn. Each week pairs one concept lesson with one hands-on investigation, and scholars keep a science notebook of their observations and drawings.

THE YEAR AT A GLANCE

1 The Nature of Science (body of knowledge)

Observing the natural world with the five senses, collaborating with a partner to gather information, keeping pictorial records, and learning that careful observation is a way to learn.

3 Physical Science (body of knowledge)

Sorting objects by properties such as size, shape, color, temperature, weight, and texture; changing the shape of materials; observing that sound comes from vibration; and how a push or pull changes the way things move.

2 Earth and Space Science (body of knowledge)

The day and night pattern, when the Sun and Moon can be seen, near and far and big and small as seen from Earth, and how gravity pulls objects toward the ground.

4 Life Science (body of knowledge)

The five senses and related body parts, observing and comparing plants and animals, and noticing that some stories show animals doing things they cannot really do.

A TYPICAL WEEK

- Two science lessons each week.
- The first introduces a core idea through reading, examples, and guided observation.
- The second is hands-on: an investigation, sorting task, or simple experiment where scholars observe, describe, and record.
- Scholars keep a science notebook of their observations and labeled drawings.

TIME COMMITMENT

160 minutes / week

two lessons a week, about 80 minutes each

- Each week pairs one concept lesson with one hands-on investigation.
- Investigations and projects may run a little longer.

WHAT SCHOLARS NEED

- A science notebook for observations and labeled drawings.
- Simple household materials for hands-on investigations.
- A computer with reliable internet and access to Canvas.
- An adult to help young scientists with hands-on activities (especially in Kindergarten).

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting twice a week on the school's weekly schedule.